

Know-value Logic with Successor Arithmetic

Hongyi Wang

Department of Philosophy, Peking University

AiML 2026

July 2nd, 2026

Table of contents

1. Background and Motivation

2. Know-value Logic with Successor Arithmetic

3. Main Results

4. Future Work

Background and Motivation

Background

Epistemic Logic $\neq$ the logic of “know-that” only.

Background

Epistemic Logic $\neq$ the logic of "know-that" only.

There are many know-wh expressions such as:

- I know whether the claim is true.
- I know what your password is.
- I know how to swim.
- I know why he was late.
- I know who proved this theorem.
- I know where she has been.

The program of studying such "know-wh" expressions was proposed by Wang (2018).

Background

Epistemic Logic $\neq$ the logic of "know-that" only.

There are many know-wh expressions such as:

- I know whether the claim is true.
- I know what your password is.
- I know how to swim.
- I know why he was late.
- I know who proved this theorem.
- I know where she has been.

The program of studying such "know-wh" expressions was proposed by Wang (2018).

Our work focuses on one specific kind of "know-what": know-value logic.

Know-value Logic

In his seminal work in 1989, Plaza proposed the public announcement logic. In the same paper, Plaza also introduced the know-value operator *Kv*, and listed a few valid formulas of **PAL** + *Kv*. And the Kv operator is used to analyze the Sum-and-Product puzzle.

Know-value Logic

In his seminal work in 1989, Plaza proposed the public announcement logic. In the same paper, Plaza also introduced the know-value operator Kv, and listed a few valid formulas of $\mathbf{PAL} + Kv$. And the Kv operator is used to analyze the Sum-and-Product puzzle.

Wang and Fan (2013) introduced the conditional Know-value operator $Kv_i(\varphi, c)$, and they provided a axiomatization for the resulting logic $\mathbf{ELKv}^r$ and also $\mathbf{PALKv}^r$ by reduction. The completeness proof of the multi-agent case was published in AIML 2014.

Know-value Logic

In his seminal work in 1989, Plaza proposed the public announcement logic. In the same paper, Plaza also introduced the know-value operator Kv, and listed a few valid formulas of $\mathbf{PAL} + Kv$. And the Kv operator is used to analyze the Sum-and-Product puzzle.

Wang and Fan (2013) introduced the conditional Know-value operator $Kv_i(\varphi, c)$, and they provided a axiomatization for the resulting logic $\mathbf{ELKv}^r$ and also $\mathbf{PALKv}^r$ by reduction. The completeness proof of the multi-agent case was published in AIML 2014.

$$!\mathrm{Kv}^r : [\phi]Kv_i(\psi, c) \leftrightarrow \big(\phi \rightarrow Kv_i\big(\phi \wedge [\phi]\psi, c\big)\big)$$

Know-Value Logic

Operator	Meaning	Logical Form	Example
$Kv_i c$	Agent i knows the value of c.	$\exists x\, K_i(c = x)$	"i knows what the password is."
$Kv_i(\phi, c)$	Agent i knows the value of c under condition ϕ.	$\exists x\, K_i(\phi \to c = x)$	"i knows the password, given that it is a 4-digit number."

Formalizing the Puzzle

The following is the Sum-and-Product puzzle. There are two integers x and y with $1 < x < y$ and $x + y \leq 100$. P knows the value of $x \times y$, and S knows the value of $x + y$. They have the following conversation:

- P: "I don't know the two numbers."
- S: "I knew that you didn't know. I don't know either."
- P: "Now I know the two numbers."
- S: "Now I know them too."

Formalizing the Puzzle

The following is the Sum-and-Product puzzle. There are two integers x and y with $1 < x < y$ and $x + y \leq 100$. P knows the value of $x \times y$, and S knows the value of $x + y$. They have the following conversation:

- P: "I don't know the two numbers."
- S: "I knew that you didn't know. I don't know either."
- P: "Now I know the two numbers."
- S: "Now I know them too."

- not only knowing the value of constants, but also terms generated by function symbols.

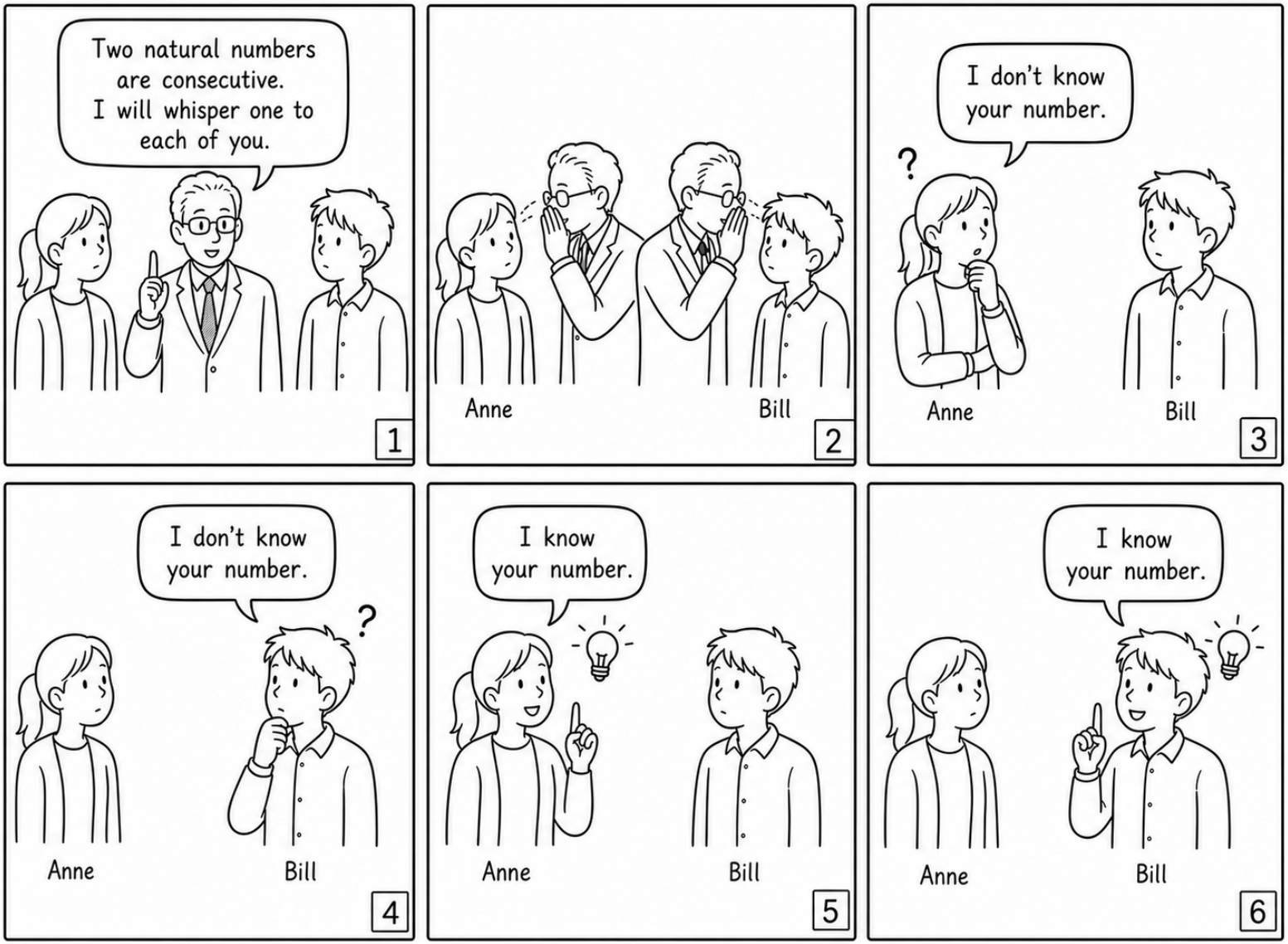
Formalizing the Puzzle

The following is the Sum-and-Product puzzle. There are two integers x and y with $1 < x < y$ and $x + y \leq 100$. P knows the value of $x \times y$, and S knows the value of $x + y$. They have the following conversation:

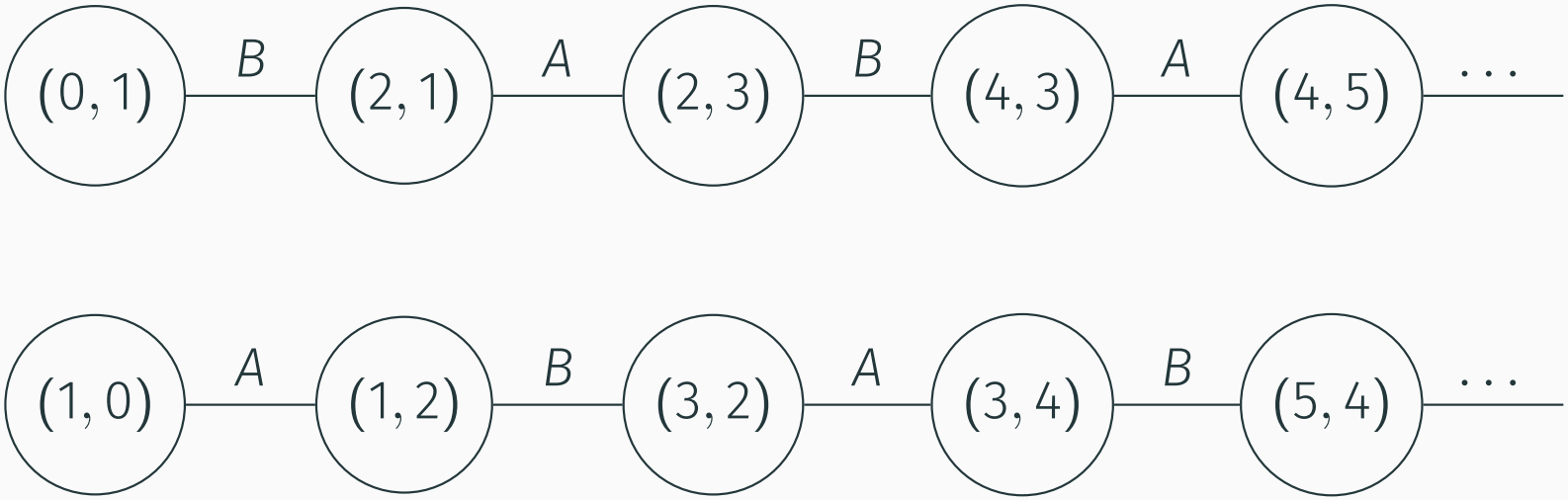
- P: "I don't know the two numbers."
- S: "I knew that you didn't know. I don't know either."
- P: "Now I know the two numbers."
- S: "Now I know them too."

- not only knowing the value of constants, but also terms generated by function symbols.
- Conbine epistemic logic with some background arithmetic theory.

Consecutive Numbers (van Ditmarsch and Kooi 2015)

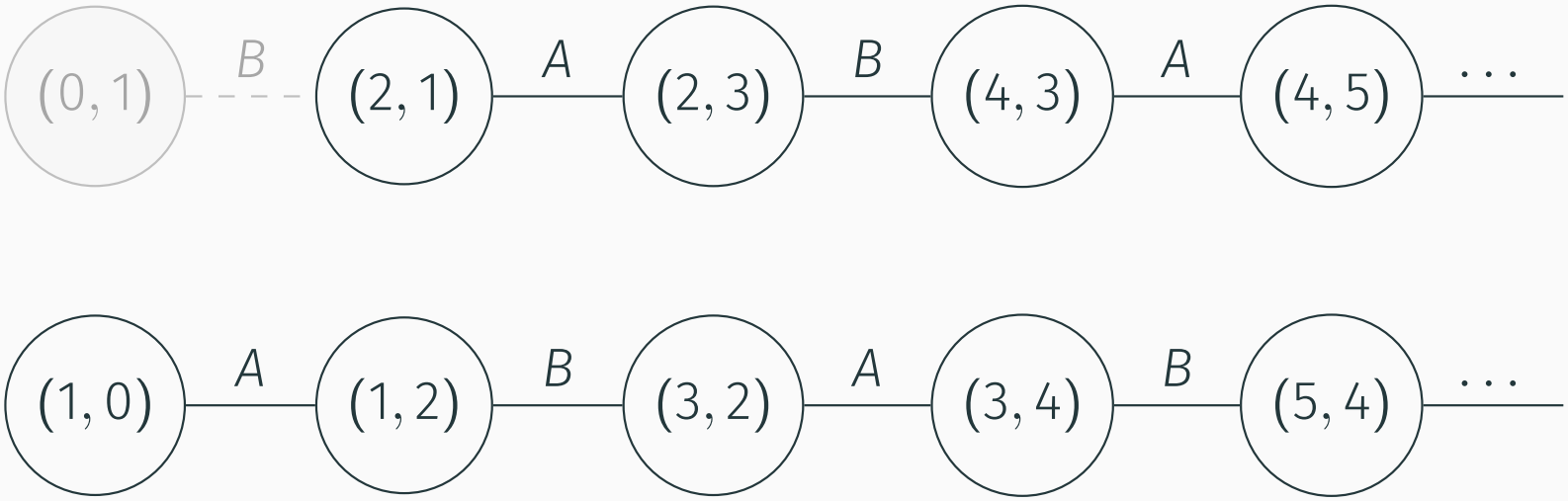


Analysis



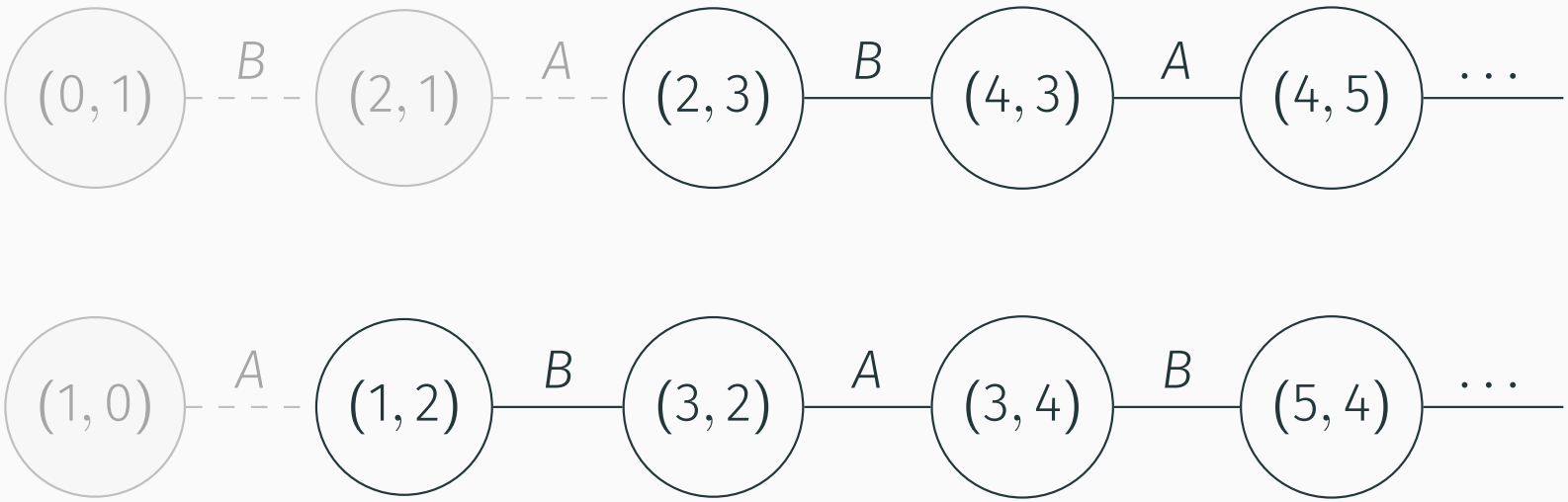
- Anne: “I don’t know your number.”

Analysis



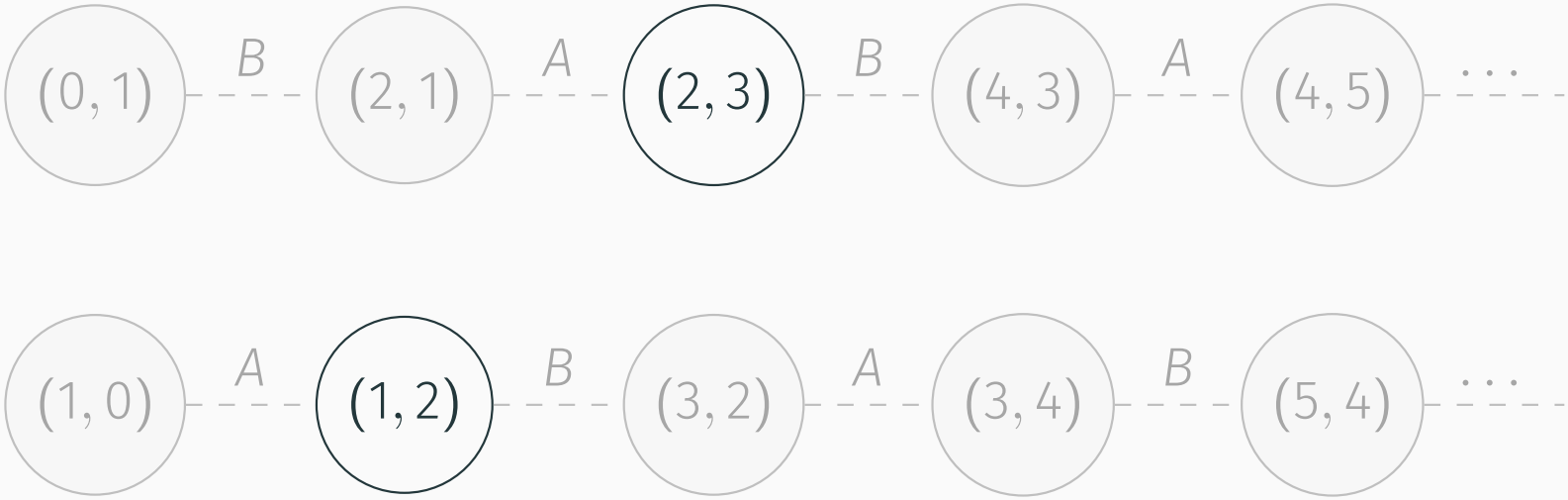
- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”

Analysis



- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”
- Anne: “I know your number.”

Analysis



- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”
- Anne: “I know your number.”
- Bill: “I know your number.”

Formalization

To formalize the puzzle, the following statements are crucial:

- Both Anne and Bill knows that their numbers are consecutive.
- If Anne's number is 0, she knows Bill's number.

Formalization

To formalize the puzzle, the following statements are crucial:

- Both Anne and Bill knows that their numbers are consecutive.
- If Anne's number is 0, she knows Bill's number.

So to formalize this puzzle, it is natural to introduce the successor function symbol S, the equality symbol $\approx$ and the constant symbol 0.

Know-value Logic with Successor Arithmetic

Language

Definition (Language PALKvSA^r(I, C))

Given a countable constant symbol set **C** and a countable set of agents **I**, the set of terms Term and formulas are defined as follows.

$$t ::= 0 \mid c \mid St$$
$$\varphi ::= \top \mid t \approx t \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid Kv_i(\varphi, t) \mid [\varphi]\varphi$$

We write $Kv_i t$ as an abbreviation of $Kv_i(\top, t)$. And $\widehat{K}_i\varphi$ is an abbreviation of $\neg K \neg \varphi$.

Language

Definition (Language PALKvSA^r(I, C))

Given a countable constant symbol set **C** and a countable set of agents **I**, the set of terms Term and formulas are defined as follows.

$$t ::= 0 \mid c \mid St$$
$$\varphi ::= \top \mid t \approx t \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid Kv_i(\varphi, t) \mid [\varphi]\varphi$$

We write $Kv_i t$ as an abbreviation of $Kv_i(\top, t)$. And $\widehat{K}_i\varphi$ is an abbreviation of $\neg K \neg \varphi$.

The announcement-free fragment is **ELKvSA**^{r}.

Semantics: Standard Models

Definition (Standard Models)

$\mathcal{M} = \langle W, \mathbb{N}, \mathsf{S}, \{\sim_i\}_{i \in \mathsf{I}}, V_{\mathsf{C}} \rangle$ is called a standard model, where $\sim_i$ is an equivalence relation on the set of possible worlds W for each agent i. Here $\mathbb{N}$ is the set of natural numbers and S is the successor function on $\mathbb{N}$, and $V_{\mathsf{C}} : \mathsf{C} \times W \to \mathbb{N}$ assigns a natural number for each constant-world pair.

Define the assignment for terms $V_{\mathrm{T}} : \mathrm{Term} \times W \to \mathbb{N}$ as follows:

$$V_{\mathrm{T}}(0, w) = \mathsf{0}$$
$$V_{\mathrm{T}}(c, w) = V_{\mathsf{C}}(c, w)$$
$$V_{\mathrm{T}}(St, w) = \mathsf{S}(V_{\mathrm{T}}(t, w))$$

Semantics

Then the satisfaction relation is defined as follows:

$$
\begin{array}{lcl}
\mathcal{M}, w \models t_1 \approx t_2 & \iff & V_{\mathrm{T}}(t_1, w) = V_{\mathrm{T}}(t_2, w) \\
\mathcal{M}, w \models Kv_i(\varphi, t) & \iff & \text{there exists an } n \in \mathbb{N}, \text{ for each } u \in W \\
 & & \text{such that } w \sim_i u \text{ and } \mathcal{M}, u \models \varphi, \\
 & & V_{\mathrm{T}}(t, u) = n. \\
\mathcal{M}, w \models [\varphi]\psi & \iff & \mathcal{M}, w \models \varphi \text{ implies } \mathcal{M}|_{\varphi}, w \models \psi
\end{array}
$$

Semantics

Then the satisfaction relation is defined as follows:

$$
\begin{array}{lcl}
\mathcal{M}, w \models t_1 \approx t_2 & \iff & V_{\mathrm{T}}(t_1, w) = V_{\mathrm{T}}(t_2, w) \\
\mathcal{M}, w \models Kv_i(\varphi, t) & \iff & \text{there exists an } n \in \mathbb{N}, \text{ for each } u \in W \\
 & & \text{such that } w \sim_i u \text{ and } \mathcal{M}, u \models \varphi, \\
 & & V_{\mathrm{T}}(t, u) = n. \\
\mathcal{M}, w \models [\varphi]\psi & \iff & \mathcal{M}, w \models \varphi \text{ implies } \mathcal{M}|_\varphi, w \models \psi
\end{array}
$$

Proposition

ELKvSA^r *is not compact w.r.t. standard models.*

Proof.

The set $\{\neg c \approx S^n 0 \mid n \in \mathbb{N}\}$ is finitely satisfiable, but the whole set is not satisfiable. □

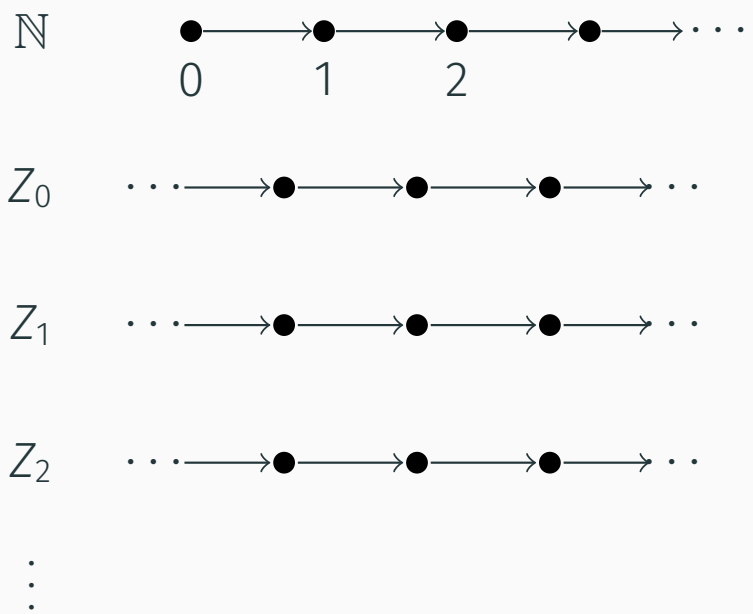
Semantics: Non-standard Models

We will define non-standard models, and prove the strong completeness w.r.t. non-standard models, and then use this result to prove the weak completeness w.r.t. standard models.

Semantics: Non-standard Models

We will define non-standard models, and prove the strong completeness w.r.t. non-standard models, and then use this result to prove the weak completeness w.r.t. standard models.

We replace $(\mathbb{N}, \mathsf{S})$ with the countably saturated model $(\mathbb{N}^*, \mathsf{S}^*)$.



Axiom System $\mathbb{ELKVSA}^r$

System $\mathbb{ELKVSA}^r$

Axiom Schemas

TAUT	all the instances of tautologies
DISTK	$K_i(\varphi \to \psi) \to (K_i\varphi \to K_i\psi)$
T	$K_i\varphi \to \varphi$
4	$K_i\varphi \to K_iK_i\varphi$
5	$\neg K_i\varphi \to K_i\neg K_i\varphi$
ID	$t \approx t$
SI	$t \approx t' \to (\alpha \to \alpha(t'//t))$
	where α is an atomic formula
SA1	$\neg 0 \approx St$
SA2	$St_1 \approx St_2 \to t_1 \approx t_2$
SA3	$\neg S^n t \approx t$, where $n \geq 1$
Kv^r0	Kv_i0
$\mathrm{Kv}^r\mathrm{S}$	$Kv_i(\varphi, t) \leftrightarrow Kv_i(\varphi, St)$
Kv^r4	$Kv_i(\varphi, t) \to K_iKv_i(\varphi, t)$
KVE1	$Kv_it_1 \wedge K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$
KVE2	$Kv_it_1 \wedge Kv_it_2 \to (K_i(t_1 \approx t_2) \vee K_i\neg(t_1 \approx t_2))$

Rules

MP	$\dfrac{\varphi \quad \varphi \to \psi}{\psi}$
NECK	$\dfrac{\varphi}{K_i\varphi}$
RE	$\dfrac{\psi \leftrightarrow \chi}{\varphi \leftrightarrow \varphi[\psi/\chi]}$
Kv^r-Elim	$\dfrac{(Kv_ic \wedge K_i(\varphi \to c \approx t)) \to \psi}{Kv_i(\varphi, t) \to \psi}$
	where c does not occur in φ, ψ, t

Intuition for the Axioms: Part 1

Axiom Schemas

TAUT	all instances of tautologies
DISTK	$K_i(\varphi \to \psi) \to (K_i\varphi \to K_i\psi)$
T	$K_i\varphi \to \varphi$
4	$K_i\varphi \to K_iK_i\varphi$
5	$\neg K_i\varphi \to K_i\neg K_i\varphi$
Kv^r4	$Kv_i(\varphi, t) \to K_iKv_i(\varphi, t)$

Kv^r5 : $\neg Kv_i(\varphi, t) \to K_i\neg Kv_i(\varphi, t)$ is provable.

Rules

MP	$\dfrac{\varphi \quad \varphi \to \psi}{\psi}$
NECK	$\dfrac{\varphi}{K_i\varphi}$
RE	$\dfrac{\psi \leftrightarrow \chi}{\varphi \leftrightarrow \varphi[\psi/\chi]}$

Intuition for the Axioms: Part 2

Equality Axioms

Axiom	Schema	Intuition
ID	$t \approx t$	Equality is reflexive.
SI	$t \approx t' \to (\alpha \to \alpha(t'//t))$ where α is atomic	Equal terms can replace each other in atomic formulas.

The symmetry and transitivity of equality are provable in $\mathbb{ELKVSA}^r$. Moreover, $t_1 \approx t_2 \to St_1 \approx St_2$ is also provable.

Intuition for the Axioms: Part 3

In $\mathbb{ELKVSA}^r$	First-order successor arithmetic
SA1: $\neg 0 \approx St$	$\forall x \neg 0 \approx Sx$
SA2: $St_1 \approx St_2 \to t_1 \approx t_2$	$\forall x \forall y \, (Sx \approx Sy \to x \approx y)$
SA3: $\neg S^n t \approx t$, where $n \geq 1$	$\forall x \neg S^n x \approx x$, for each $n \geq 1$
This principle cannot be expressed in our language, since it has no quantifiers.	$\forall x \, (\neg x \approx 0 \to \exists y \, x \approx Sy)$.

Intuition for the Axioms: Part 4

Axiom		Intuition
$\mathrm{Kv}^r 0$	$Kv_i 0$	0 is a rigid constant.
$\mathrm{Kv}^r \mathrm{S}$	$Kv_i(\varphi, t) \leftrightarrow Kv_i(\varphi, St)$	S is a rigid injection.

The axiom $\mathrm{Kv}^r \mathrm{S}$ plays a crucial role in our completeness proof.

Intuition for the Axioms: Part 5

KVE1 $\quad Kv_i t_1 \wedge K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$

KVE2 $\quad Kv_i t_1 \wedge Kv_i t_2 \to (K_i t_1 \approx t_2 \vee K_i \neg t_1 \approx t_2)$

Kv^r-Elim $\quad \dfrac{\vdash (Kv_i c \wedge K_i(\varphi \to c \approx t)) \to \psi}{\vdash Kv_i(\varphi, t) \to \psi}$

where c does not occur in φ, ψ, t

Mention-some reading	Mention-all reading
$\exists x\, K_i(\varphi \to c \approx x)$	$\forall x\, (K_i(c \approx x) \vee K_i \neg(c \approx x))$
KVE1 $\exists$-Intro	KVE2 $\forall$-Elim
Kv^r-Elim $\exists$-Elim	

Power of Kv^r-Elim

In Wang and Fan's system, since equality is not in the language, the following three formulas are axioms. But in our system, they become provable.

1. $\mathsf{DISTKv}^r : K_i(\varphi \to \psi) \to (Kv_i(\psi, t) \to Kv_i(\varphi, t))$
2. $\mathsf{Kv}^r\bot : Kv_i(\bot, t)$
3. $\mathsf{Kv}^r\vee : \widehat{K}_i(\varphi \wedge \psi) \wedge Kv_i(\varphi, t) \wedge Kv_i(\psi, t) \to Kv_i(\varphi \vee \psi, t)$

Two Useful Theorems

The following schemas are provable in $\mathbb{ELKVSA}^r$:

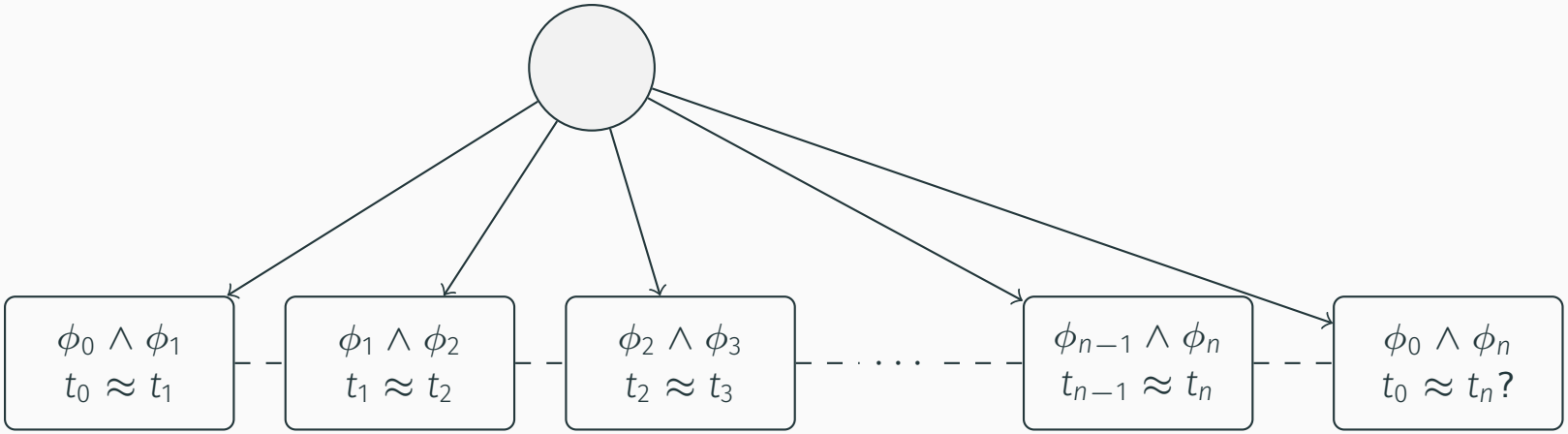
- GKVE :

$$\left(\bigwedge_{k=0}^{n} Kv_i(\phi_k, t_k)\right) \wedge \left(\bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})\right) \to K_i(\phi_0 \wedge \phi_n \to t_0 \approx t_n).$$

- GKv$^r\vee$:

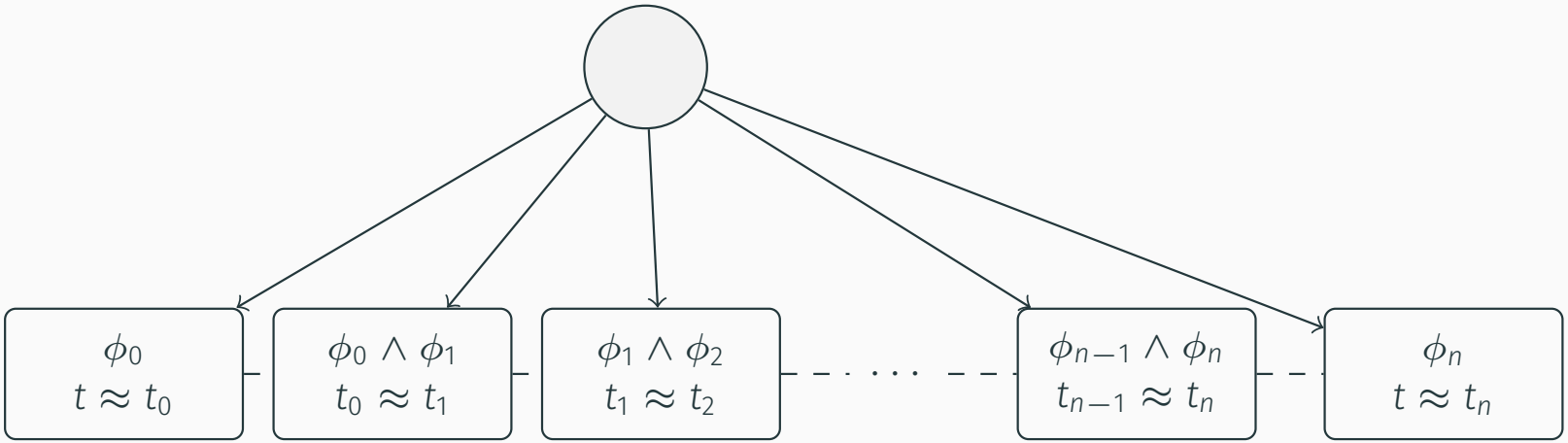
$$\left(\bigwedge_{k=0}^{n} Kv_i(\phi_k, t_k)\right) \wedge \left(\bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})\right) \to Kv_i\left(\bigvee_{k=0}^{n}(\phi_k \wedge t \approx t_k), t\right).$$

Why GKVE is Valid



$$\left(\bigwedge_{k=0}^{n} Kv_i(\phi_k, t_k)\right) \wedge \left(\bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})\right) \to K_i(\phi_0 \wedge \phi_n \to t_0 \approx t_n).$$

Why GKv$^r\vee$ is Valid



$$\left(\bigwedge_{k=0}^{n} Kv_i(\phi_k, t_k)\right) \wedge \left(\bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})\right) \to Kv_i\left(\bigvee_{k=0}^{n} (\phi_k \wedge t \approx t_k), t\right).$$

Main Results

Main Results

- We first prove the strong completeness of $\mathbb{ELKVSA}^r$ w.r.t. non-standard models.

Main Results

- We first prove the strong completeness of $\mathbb{ELKVSA}^r$ w.r.t. non-standard models.
- Then we show that if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.

Main Results

- We first prove the strong completeness of $\mathbb{ELKVSA}^r$ w.r.t. non-standard models.
- Then we show that if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.
- Finally, we show that if ϕ is satisfiable in a finite non-standard model, then we can transform it into a finite standard model.

Main Results

- We first prove the strong completeness of $\mathbb{ELKVSA}^r$ w.r.t. non-standard models.
- Then we show that if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.
- Finally, we show that if ϕ is satisfiable in a finite non-standard model, then we can transform it into a finite standard model.
- Thus, we can conclude that $\mathbb{ELKVSA}^r$ is weakly complete w.r.t. standard models. Moreover, **ELKvSA**^{r} is decidable.

Strong Completeness: Tree Unraveling and Valuation

1. First consider the canonical frame from Wang and Fan (2013), which are used to prove the completeness for the single-agent case.

Strong Completeness: Tree Unraveling and Valuation

1. First consider the canonical frame from Wang and Fan (2013), which are used to prove the completeness for the single-agent case.
2. Unravel the frame into a tree-like frame and take the S5 closure.

Strong Completeness: Tree Unraveling and Valuation

1. First consider the canonical frame from Wang and Fan (2013), which are used to prove the completeness for the single-agent case.
2. Unravel the frame into a tree-like frame and take the S5 closure.
3. Construct a valuation over the non-standard arithmetic domain.

Why Tree Unraveling Necessary

We give an example:

$$\{Kv_1c, \neg Kv_1d, Kv_2d, \neg Kv_2c, c \approx d\}$$

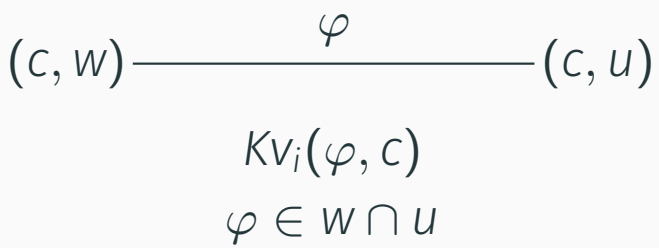
$$1 \mid 2$$

$$\{Kv_1c, \neg Kv_1d, Kv_2d, \neg Kv_2c, \neg c \approx d\}$$

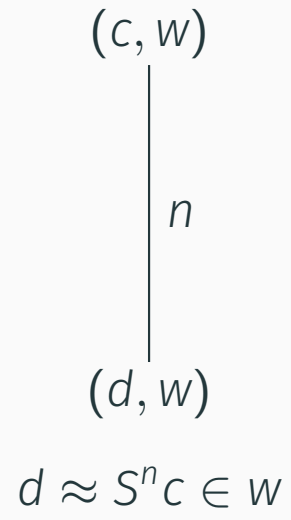
We cannot give a valuation! So tree-unraveling is needed.

Strong Completeness: Valuation

We assign values to constant-world pairs. The valuation is required to satisfy exactly the following two kinds of constraints:



Kv-constraints



Arithmetic constraints

We impose all and only these constraints. For every $\neg Kv_i(\varphi, c) \in w$, the construction and the axioms ensure that there are two accessible φ-successors which are not forced equal by these constraints.

FMP: Tree Unraveling and Selection

- Unravel the model into a tree-like model.
- Let $D = \mathrm{depth}(\phi)$. We only keep worlds that can still affect the truth of ϕ within modal depth D.
- If a selected world satisfies $\neg K_i \psi$, select one accessible witness where ψ is false.
- If a selected world satisfies $\neg Kv_i(\chi, t)$, select two accessible χ-worlds where t has different values.
- The selected submodel is finite, and truth of the relevant subformulas is preserved by induction on modal depth.

From Non-standard Models to Standard Models

Suppose $\mathcal{N} = \langle W, \mathbb{N}^\star, \mathsf{S}^\star, \{\sim_i\}_{i\in I}, V^\star_{\mathsf{C}}\rangle$ is a finite non-standard model satisfying ϕ. Only finitely many values are relevant:

$$U = \{V^\star_{\mathsf{C}}(c, w) \mid c \text{ occurs in } \phi,\ w \in W\} \cup \{0\}.$$

Since both W and the set of constants occurring in ϕ are finite, U is finite.

From Non-standard Models to Standard Models

Suppose $\mathcal{N} = \langle W, \mathbb{N}^\star, \mathsf{S}^\star, \{\sim_i\}_{i\in I}, V_{\mathsf{C}}^\star\rangle$ is a finite non-standard model satisfying ϕ. Only finitely many values are relevant:

$$U = \{V_{\mathsf{C}}^\star(c, w) \mid c \text{ occurs in } \phi,\ w \in W\} \cup \{0\}.$$

Since both W and the set of constants occurring in ϕ are finite, U is finite.

We construct an injection $f : U \to \mathbb{N}$ such that:

- $f(x) = x$ for $x \in U \cap \mathbb{N}$;

From Non-standard Models to Standard Models

Suppose $\mathcal{N} = \langle W, \mathbb{N}^\star, \mathsf{S}^\star, \{\sim_i\}_{i\in I}, V_{\mathsf{C}}^\star\rangle$ is a finite non-standard model satisfying ϕ. Only finitely many values are relevant:

$$U = \{V_{\mathsf{C}}^\star(c, w) \mid c \text{ occurs in } \phi,\ w \in W\} \cup \{0\}.$$

Since both W and the set of constants occurring in ϕ are finite, U is finite.

We construct an injection $f : U \to \mathbb{N}$ such that:

- $f(x) = x$ for $x \in U \cap \mathbb{N}$;
- For $x, y \in U$ from the same component, $f(x) - f(y) = x - y$;

From Non-standard Models to Standard Models

Suppose $\mathcal{N} = \langle W, \mathbb{N}^\star, \mathsf{S}^\star, \{\sim_i\}_{i\in I}, V_{\mathsf{C}}^\star\rangle$ is a finite non-standard model satisfying ϕ. Only finitely many values are relevant:

$$U = \{V_{\mathsf{C}}^\star(c, w) \mid c \text{ occurs in } \phi,\ w \in W\} \cup \{0\}.$$

Since both W and the set of constants occurring in ϕ are finite, U is finite.

We construct an injection $f : U \to \mathbb{N}$ such that:

- $f(x) = x$ for $x \in U \cap \mathbb{N}$;
- For $x, y \in U$ from the same component, $f(x) - f(y) = x - y$;
- For $x, y \in U$ from different components, $|f(x) - f(y)|$ is large enough so that no atomic formula in ϕ can tell the difference.

Conclusion

Theorem

$\mathbb{ELKVSA}^r$ *is sound and strongly complete w.r.t. non-standard models, and weakly complete w.r.t. standard models.*

Theorem

ELKvSA^r *has the finite model property w.r.t. both non-standard models and standard models.*

Theorem

ELKvSA^r *is decidable.*

Future Work

Future Work

- Consider other frame classes except S5.
- Is **ELKvSA**^{r} in PSPACE?
- Consider *KvP*: knowing the value of a predicate.
- Generalize to other theories.
- Completely formalize the “Sum and Product” puzzle.

Thanks!